SAI SANTOSH CHEETHIRALA

santoshcheethirala.me@gmail.com | +91 83281 69640 | linkedin.com/in/santoshcheethirala | github.com/santoshcheethiralame-dot | Bengaluru, India

Engineer who breaks systems to find where their guarantees end. Showed a shipping distributed database losing acknowledged writes, detects coordinated mule rings on payment graphs, and builds adversarial benchmarks designed to defeat their own thesis.

EXPERIENCE

PESU Intelligence Labs — Research Intern, RAG & LLM Reliability **Bengaluru | Jun 2026 – Jul 2026**

No Single Culprit — Causal Attribution Benchmark & Evaluation Harness for Multi-Hop RAG **github.com/santoshcheethiralame-dot/LINEUP**

Python, pytest, GitHub Actions (CI/CD), REST API, Streamlit, vLLM, BM25, PyTorch, HuggingFace Transformers, Conformal Prediction

- Built and shipped a 7.2K-LOC evaluation service — public REST API, pooled leaderboard, and Streamlit explorer — with 135 automated tests gating every change on GitHub Actions CI
- Designed a non-circular counterfactual oracle producing leak-free ground-truth labels without ever consulting the method under test, replacing an evaluation loop that had been silently circular

PROJECTS

highwater — Jepsen-Style Durability Analysis of Couchbase **github.com/santoshcheethiralame-dot/HighWater**

Go, Couchbase DCP, Porcupine linearizability checker, gocbcore patch, fault injection

- Ran a Jepsen-style durability analysis of Couchbase — one of the few major distributed databases with no published Jepsen report — showing 56 acknowledged writes lost (50 regressed, 6 absent) at durability level "majority" when the in-memory majority is killed before the disk flush completes, with the history proven non-linearizable
- Built a white-box harness that taps the DCP stream to observe prepare/commit/abort directly, rather than inferring from client responses, so "never prepared" and "prepared then aborted" are distinguishable; required patching gocbcore to expose the stream
- Confirmed the result under two independent methods that agree on both arms — a direct value comparison and Porcupine, a general linearizability checker with no knowledge of durability levels — and showed the control arm (persistToMajority) loses zero writes and stays linearizable

Untangle — Graph-Native Detection of Mule-Account Fraud Rings (UPI-style payments) **github.com/santoshcheethiralame-dot/UNTANGLE**

PyTorch Geometric, GraphSAGE, GAT, synthetic data generation, evaluation harness

- Modelled UPI-style payments as a single graph to detect coordinated mule-account rings, on the thesis that no individual hop in a laundering ring looks abnormal — which is the design of the crime and the reason per-transaction scoring cannot see it
- Built the synthetic generator adversarially against its own thesis: half of each ring is rented rather than fresh, so those mules are statistically identical to ordinary customers on every profile feature (account age log 6.82 vs 6.86), with tests asserting this parity holds — preventing a benchmark a decision tree could win

tessera — Distributed, Sharded, Replicated Key/Value Store (Rust, std-only) **github.com/santoshcheethiralame-dot/TESSERA**

Rust (0 external crates), Raft, LSM-tree, Deterministic Simulation Testing, Linearizability Checking, MVCC

- Caught a real split-brain bug via seed 99 — a duplicated vote reply let a candidate win a sub-majority, producing two leaders in one term — under deterministic simulation testing that replays any failure exactly from its seed

TECHNICAL SKILLS

Security & Assurance: Fault injection, adversarial testing, threat-model-driven benchmark design, graph anomaly detection, sandboxed execution, safety gating

Languages: Python, C, Rust, TypeScript, JavaScript, SQL, WGSL

Systems & Backend: Distributed systems (Raft, LSM storage, MVCC), FastAPI, REST APIs, Docker, Linux, WebGPU/WGSL, vLLM

Testing & Reliability: Deterministic simulation testing, randomized fault injection, linearizability checking, regression testing, pytest, CI/CD (GitHub Actions), model evaluation

EDUCATION

PES University — B.Tech, Computer Science Engineering **Bengaluru | Sep 2024 – May 2028 (Expected)**

CGPA 7.95/10 (through Sem 4) — CNR Merit Scholarship (top 25% cohort), Sem 1 — Academic Distinction Award (SGPA 7.75+), Sem 2 & Sem 4

Relevant Coursework: Data Structures & Algorithms, Design & Analysis of Algorithms, Operating Systems, Computer Networks, Machine Learning, Deep Learning, NLP, Linear Algebra, Data Science

LEADERSHIP & ACTIVITIES

Sahay AI — Student AI Club, PES University — Multimedia Domain Head **Oct 2025 – Mar 2026**

- Built a content pipeline and posting schedule for a 50+ member AI club, driving 5-6x growth in content views over three months
- National-Level Air Rifle Competitor - 10m Air Rifle, National School Games (a precision event built on sustained focus)